



An improved minimal path selection approach with new strategies for pavement crack segmentation

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ABSTRACT

Intelligent pavement detection technology provides a new research idea for identifying pavement cracks. To solve the problem that traditional minimal cost path selection (MPS) is computationally intensive and inefficient in crack segmentation, this study proposes an improved minimal cost path algorithm (IMPS). The contribution of IMPS is reflected in its more efficient path planning than MPS and better choice of source and destination of each path by using new strategies. First, a grayscale pavement image is divided into nonoverlapping cells with a certain size. Next, the darkest points in each cell are selected as candidate points and adjacent candidate points are connected based on IMPS. Then pseudocracks are removed based on postprocessing steps before cracks are well segmented. IMPS is compared with other crack segmentation algorithms to prove its reliability. The efficiency of IMPS is increased by approximately 56% than MPS. The accuracy of IMPS is higher than that of MPS and other methods, with Mean Intersection over Union reaching 73.99%.

1. Introduction

Highways and pavement facilities have made great contributions to communication between different regions and have effectively promoted the economic development of countries worldwide. Pavement suffers from different types and degrees of distress during its service period because of the regional climatic environment and exposure to repeated loads. Timely pavement maintenance effectively prevents further deterioration of minor pavement distress and decreases economic costs during maintenance, which results in better sustainability of the pavement structure. Conventional pavement distress detection relies on civil engineers visually inspecting the pavement conditions on site and then evaluating the conditions. Manual detection has the disadvantages of heavy workloads, low efficiency, strong subjectivity, and poor repeatability [1]. In addition, manual workers are unable to perform this work in risky situations, which highlights the significance and urgency of the automatic detection of pavement distress. Civil engineers and researchers worldwide are very interested in semiautomatic or automatic pavement distress detection technologies, such as ultrasonic technology, ground-penetrating radar technology, image processing technology and computer vision technology [2].

In recent decades, pavement defect identification, including the identification of raveling [3], potholes [4,5], bugholes [6], cracks [7], inlaid patches [8], and pavement markings [9], has been extensively and deeply studied. The crack recognition technologies for structure defects include thresholds [10,11], edge detection [12], wavelet transforms [13], and emerging technologies such as deep learning [14–16]. Threshold-based segmentation, as a conventional image processing technology, is widely used in pavement crack feature extraction and segmentation. Pavement cracks are extracted from complicated background regions by setting the local grayscale, global grayscale or maximum entropy threshold [17]. However, the threshold is a critical factor that can determine segmentation performance. Several simple and effective edge detection operators have been proposed. The segmentation performance for linear cracks is excellent, and the segmentation performance for complex cracks is unsatisfactory. In addition, a large amount of texture noise on asphalt pavement decreases edge detection performance. Defect detection technology based on wavelet multiscale transformation [18,19] utilizes the corresponding relationship among different frequency components, such as image defects, noise and texture and wavelet subbands, and realizes defect detection by analyzing wavelet decomposition coefficients. With the development of

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computer vision technology and image processing technology, artificial intelligence-based detection technologies have been extensively employed in structural damage detection [20], pavement defect detection [21] and bridge defect detection [22] in civil engineering. Deep learning is applied on many aspects like crack classification [23], detection [24] and segmentation [25]. Liu and his group [26] proposed a CNN-based deep learning method for crack segmentation called DeepCrack and established a benchmark dataset. On this dataset, DeepCrack performs well and can get 86.5% in F1, which reaches a high level. However, collecting crack images and labeling them can be time-consuming. A large number of manually marked defect images are needed to train and verify segmentation models, which leads to enormous labor costs. What's more, various deep learning algorithms are basically learning and testing on their own datasets, and there is a lack of a rich and diverse crack image database to uniformly measure their accuracy. Thus, we need a better method for quick crack labeling.

With the development of partial differential equations and graph theory, the minimal energy functional method and minimal cost path method [27,28], which achieve better extraction performance than other conventional methods, can be employed to detect defects. For simple linear structures, such as blood vessels, researchers have proposed methods that manually insert critical points [29] or add orientation information [30] to automatically obtain critical points to realize iterative minimal cost path search technology [31]. Pavement crack extraction algorithms based on the minimal spanning tree of target points and the combination of seed growth and minimal cost paths have been employed to extract asphalt pavement crack defects and quantify the characteristics of cracks. Several types of prior knowledge, including the source and destination of the minimal cost path, need to be predetermined [32], which decreases the efficiency of multibranch curve structures with complex topological structures, such as alligator cracks. A method for detecting the minimum energy functional iteration key points has been proposed to improve the efficiency and accuracy [33]. The Euler distance constraint relation of the minimal cost path is used to determine the stop qualification for iterations. Compared with the current image processing algorithm based on minimal paths, the above algorithm extracts curve-like objects accurately from images with less prior knowledge. However, the selection of crack seed points is still supervised.

In this article, novel search strategies of minimal cost paths are proposed to detect pavement crack defects based on image processing techniques and computer vision techniques. The proposed search strategies are insensitive to pavement markings and other external noise since they are based on the minimal cost path search and not grayscale thresholds. The strategies are applied to extract crack defects from 2D images under different pavement conditions. The experimental results show that the segmentation efficiency of the improved minimal cost path is better than that of the conventional minimal cost path, and the segmentation performance is also better than MPS and other conventional segmentation techniques such as Otsu method, region growing and edge detection. The contributions of this article are summarized as follows:

- (a) New strategies are proposed to determine the source and destination points of the minimal cost path and to address the search regions and eliminate overlapping paths. These strategies effectively settle the problem of the low computational efficiency of the conventional minimal cost path method.
- (b) The improved minimal cost path method has a rather stable segmentation performance for pavement markings, shadows and other noises.

The remainder of this paper is organized as follows: Denoising methods are evaluated and chosen in section 2.1. To improve computational efficiency and segmentation performance, novel search strategies are proposed and introduced in section 2.2 in detail. The

experimental results are presented in section 3. Section 4 discusses the advancements of the proposed method. Section 5 analyses the future of IMPS. The paper concludes with some remarks in section 6.

2. Methodology

2.1. Pavement image denoising

Pavement image noise is divided into two categories. One category is the intrinsic noise caused by the image capture system. This kind of noise cannot be avoided and has the characteristics of isolation and discontinuity. The other category is the external noise caused by the working environment, such as sunny or cloudy illumination, and the characteristics of the asphalt pavement surface, such as the aggregate texture of the asphalt pavement, cuts on cement pavement, pavement markings and tire marks. The aim of pavement crack image segmentation is to accurately extract crack regions or crack pixels from complex background regions; quantify the crack characteristics, such as the length, area, and average width; and classify crack categories to provide guidance for pavement maintenance departments to prepare maintenance budgets. Since isolated noise points can be detected as candidate crack points for minimal cost path segmentation and impact segmentation accuracy, pavement images are first denoised to filter out isolated noise points. The nonlocal means (NLM) filtering method is used to filter out isolated noise in this paper.

The main idea of nonlocal means denoising is that the gray level of a pixel at a certain position is evaluated by the weighted average of the corresponding pixel with a similar neighborhood structure in the image. Nonlocal means denoising adopts a nonlocal correlation and pixel patches to evaluate photometric similarity [34,35], which retains more edge details. The nonlocal means filtering procedures are introduced in detail in [36]. Although nonlocal means filtering has been verified to achieve a good denoising performance [37], it has not been applied to the field of pavement engineering. To verify the performance of nonlocal means denoising for asphalt pavement distress images, the peak signal-to-noise ratio (PSNR, unit: dB) and the structural similarity index (SSIM) are employed to illustrate robustness. The PSNR metric is defined by the mean squared error (MSE) metric. For two 2-D grayscale images $I(i,j)$ and $J(i,j)$ of size $m \times n$, the MSE is defined as follows:

$$MSE = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i,j) - J(i,j)]^2 \quad (1)$$

where (i,j) represents the pixel position in the image. The PSNR is defined as follows:

$$PSNR = 10 \cdot \lg \left(\frac{MAX_I^2}{MSE} \right) = 20 \cdot \lg \left(\frac{MAX_I}{\sqrt{MSE}} \right) \quad (2)$$

where the image used in this paper is an 8-bit grayscale image; thus, MAX_I is $2^8 - 1 = 255$, which is 1 after normalization.

The SSIM is another index that is used to evaluate denoising performance. The SSIM comprehensively considers the similarity of the luminance, contrast and structure of images [37] and is defined as follows.

$$SSIM(I, J) = l(I, J) + c(I, J) + s(I, J) \quad (3)$$

where $l(I, J)$, $c(I, J)$, and $s(I, J)$ obey the following equation:

$$\begin{aligned} l(I, J) &= \frac{2\mu_I\mu_J + C_1}{\mu_I^2 + \mu_J^2 + C_1} \\ c(I, J) &= \frac{2\sigma_I\sigma_J + C_2}{\sigma_I^2 + \sigma_J^2 + C_2} \\ s(I, J) &= \frac{\sigma_{IJ} + C_3}{\sigma_I\sigma_J + C_3} \end{aligned} \quad (4)$$

where $C_1 = (K_1 \times L)^2$, $C_2 = (K_2 \times L)^2$, and $C_3 = C_2/2$. Generally, let K_1

$= 0.01$, $K_2 = 0.03$, and $L = 255$. μ_I and μ_J are the mean gray levels of images I and J , respectively; σ_I and σ_J are the variances of the gray levels, respectively; and σ_{IJ} is the covariance of the gray levels.

2.2. Minimal cost path segmentation

2.2.1. Cell images

The conventional minimal path algorithm such as Dijkstra algorithm needs to traverse all the pixel points of a grayscale image to obtain the minimal path from the source point to the destination point. Traversing all the pixels of an image implies occupying more computational resources and increasing time consumption, which are not acceptable for the fast and high-efficiency identification of crack targets. A reliable method is to first decrease the computational cost to extract potential crack pixel points and then connect them according to certain rules. In this paper, the converted grayscale image is divided into subblocks with a size of $P \times P$ (P can be any integer such as 4, 8, 10, 16, 20, 40, etc.), and each subblock image is called a cell. The point with the smallest grayscale value in a cell is selected as the crack candidate point of the cell since the crack pixels have lower grayscale values than the background pixels. However, the crack region is only a small part of the image, which shows that not all of the darkest pixels in cells belong to the crack region. To address this problem, a threshold based solution for removing noncrack pixels is proposed. The method is used to remove candidate points with grayscale values greater than a threshold T_g in the crack candidate point set. The threshold obeys (5).

$$T_g = \mu_g - k_g \sigma_g \quad (5)$$

where μ_g and σ_g are the mean and standard deviation of the grayscale of the candidate point set, respectively, and k_g is the empirical

coefficient. The crack candidate points are shown in Fig. 1.

Since isolated noise points may be wrongly chosen as candidate crack points for minimal cost path segmentation and impact the segmentation accuracy, the pavement images are first denoised to filter out isolated noise points. Nonlocal means filtering method is used for this purpose. The denoising procedures were elaborated by Liang et al. [25]. The reason for choosing nonlocal means as the preprocessing method is explained in detail in section 4.

2.2.2. Minimal cost path set

Although a set of crack candidate points has been determined, the candidate points are isolated and have no connection. Therefore, the minimal cost path method is applied to connect them. The principle of the minimal cost path based on the Dijkstra algorithm is introduced in this section. First, two empty sets V_1 and V_2 are established. The points determined to belong to the minimal cost path are placed in set V_1 , and the remaining points are placed in set V_2 . First, set V_1 contains only the source point s , and whether all the remaining points are on the minimal cost path is not determined. The grayscale of each pixel is defined as its cost. The points in set V_2 are individually added to V_1 in increasing order of the minimal cost path until destination d is added to set V_2 . This process is called vertex iteration. When vertex iteration is completed, all the points belonging to the minimal cost path between source s and destination d are placed in set V_1 , and the minimal cost path can be determined. The path cost function $C_{s,d}$ is defined as the mean grayscale of the pixel points on the path between s and d , which obeys (6).

$$C_{s,d} = \frac{\sum_{p \in V_1} \text{Gray}(p)}{\sum_{p \in V_1} n(p)} \quad (6)$$

where $\text{Gray}(p)$ denotes the grayscale of any point p in set V_1 and $n(p)$

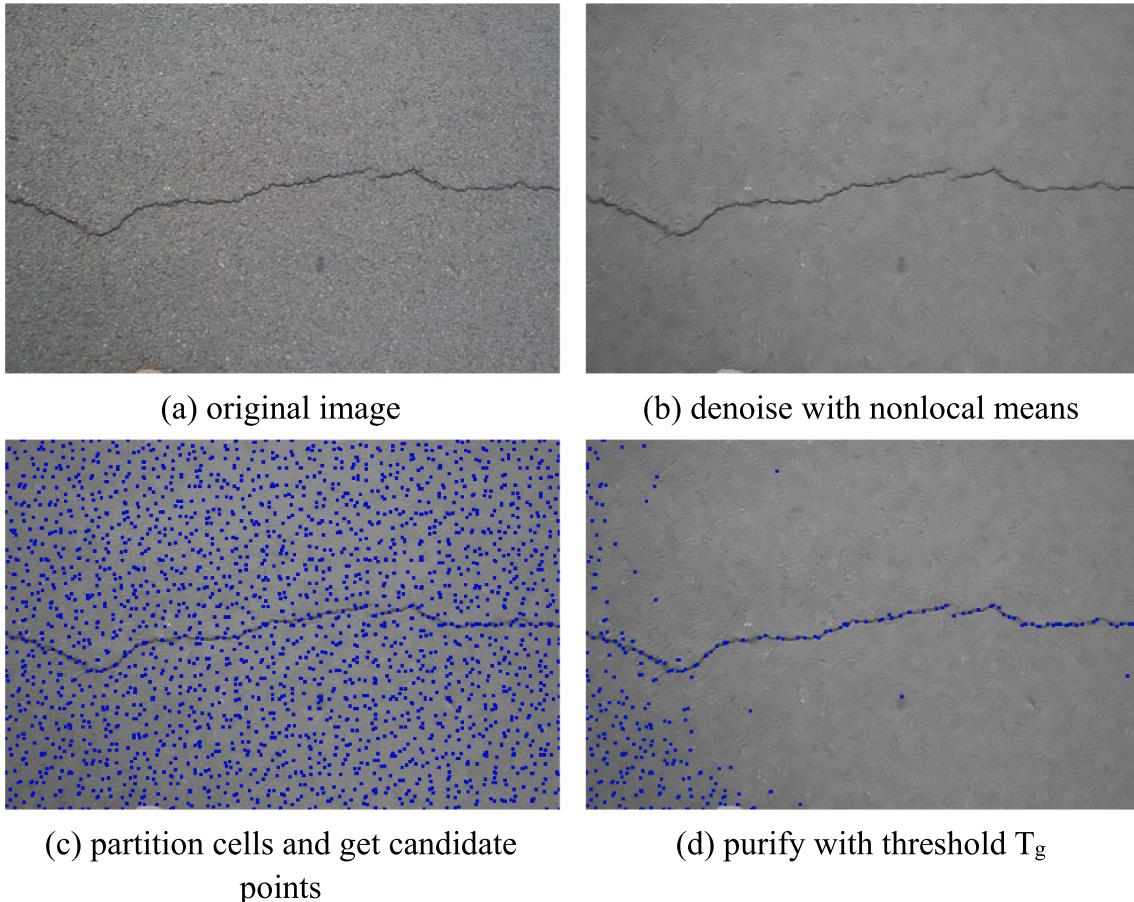


Fig. 1. Crack candidate point set.

represents the number of pixels in set V_1 . A simple graphic of the procedure illustrates the calculation process for the minimal cost path, as shown in Fig. 2.

2.2.3. Optimized path searching strategy

To address the matter of source and destination selection and minimal cost path planning in the crack extraction process, a novel cost path searching strategy was proposed to quickly and efficiently select the sources and plan the paths. Since it is not determined which of the two crack candidate points in a pair of adjacent cells is the source and which is the destination, the path connecting them will be detected more than once. To address the problem of repetitive paths, an innovative method was proposed to determine adjacent cells and distinguish between the source and destination. For any cell C , the top-right, right, bottom-right and lower cells around it are defined as its adjacent cells marked as TR , RT , BR and BW , respectively, as shown in Fig. 3. Fig. 4 illustrates the selection strategy of source and destination points more clearly, and the specific procedure is as follows.

- The top-left cell of the original image is set as the current cell C .
- If there is a crack candidate point in cell C , then the candidate point is set as the source. If there is no candidate point in cell C , then the current cell is moved to the right (or down if it is already at the rightmost end) to set a new current cell C until a candidate point is included in the current cell C .
- If the current cell C has no less than one adjacent cell, then perform step (c); otherwise, the new current cell C is set as step (a).
- Determine whether crack candidate points exist in cell C 's adjacent cells. If yes, then these candidate points are set as the destination points of the source point in cell C . If no, then the crack candidate point in cell C is removed from the candidate point set.
- Procedures (b) to (d) are repeatedly performed, and the search for the source and destination is performed until all the cells in the image are set as cell C once.

After the source and destination candidate points are determined by the above search strategy, it is necessary to find the minimal cost path connecting the source and destination points on the pixel scale. In the process of searching for the minimal cost path between source and destination points, we use the 4-neighborhood to define the neighboring points of a pixel point, which means that the left, right, up and down pixels are defined as the neighboring pixels of the current pixel point.

Then, we start searching for the minimal cost path in the search region, which is defined as small as possible but still large enough to include the source and destination points, thus improving the Dijkstra algorithm to find the minimal path efficiently and shorten the calculation time.

Taking Fig. 3 as an example, it's assumed that the candidate point in cell C is the source point and the candidate point in cell TR is the destination point of the minimal cost path. Then, the upper boundary and right boundary of cell TR and the left boundary and lower boundary of cell C are used to limit the search region, as the red frame shows. Similarly, the minimal cost path search region for the destination point in cell RT is the blue frame. Cell BR and cell BW match the yellow frame and the green frame, respectively. Then, the implementation procedures of searching minimal cost path based on the Dijkstra algorithm [38] are as follows:

- Two sets, V_1 and V_2 , are created to store the investigated pixels and unexplored pixels, respectively. When initialized, V_1 contains only the source point s , which is set as the current point p , while the remaining pixels belong to V_2 . The parent nodes are stored in set PN .
- The cost of path $C_{s,p(i)}$ is calculated from the source s to each adjacent point $p(i)$ of the current point p , where $i = 1, \dots, m$, $m \leq 4$. $p(i)$ is included in set V_2 .
- The certain $p(i)$ that minimizes the cost of the current path is found, moved out of V_2 and added to V_1 . p is set as the parent node of this point $p(i)$, and p is added to PN . Then, $p(i)$ is set as the new current point p .
- Steps (b) to (c) are repeated until current point p coincides with destination point d , and then, the cycle is stopped.
- Destination point d is traced back to source point s to form a path by using their parent nodes, and this path is the minimal cost path between s and d .

It is worth mentioning that the traditional Dijkstra algorithm needs to obtain the minimal path after traversing all the points in the search region until all the points in V_2 are moved to V_1 . In this paper, if the current point p coincides with the destination point d , then the path finding process is terminated. It is considered that the backtracking path is the minimal path between the source and destination points, and there is no need to further traverse all the points. The specific process is shown in Fig. 5. This improvement can increase the efficiency of the algorithm and reduce the operating time. After finding all the minimal cost paths between every pair of adjacent source points and destination points, the minimal cost path set (MCPS) can be established.

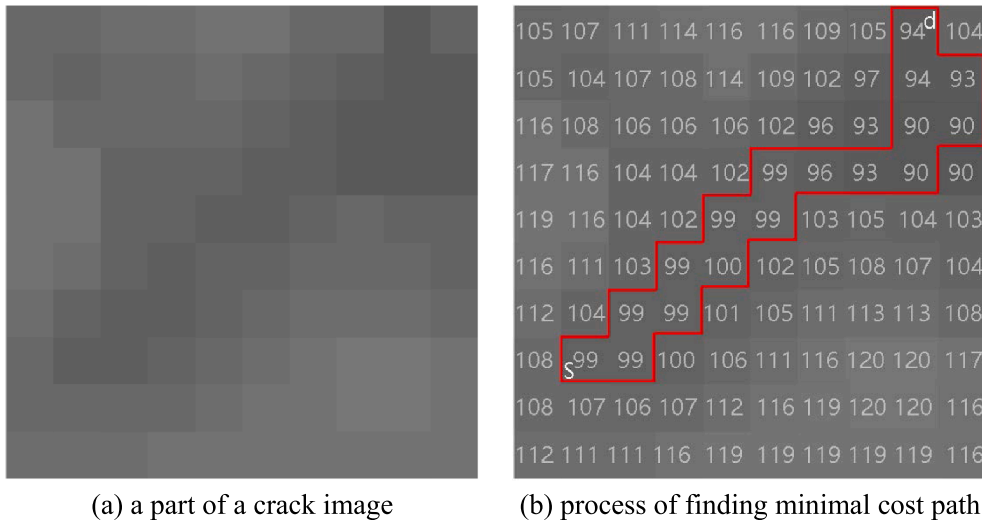


Fig. 2. Marked grayscale map with the minimal cost path on it.

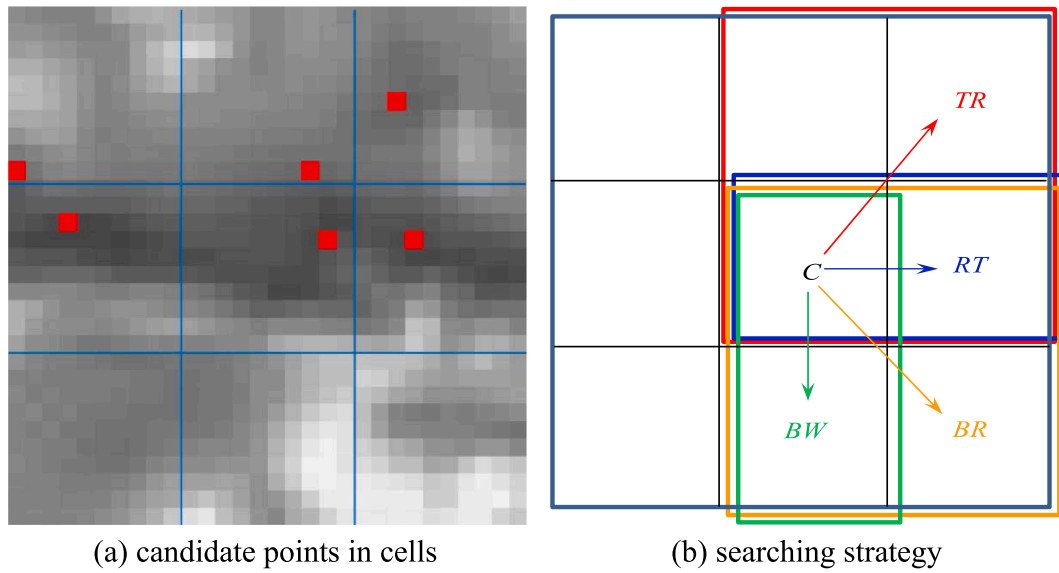


Fig. 3. Search region of the minimal cost path.

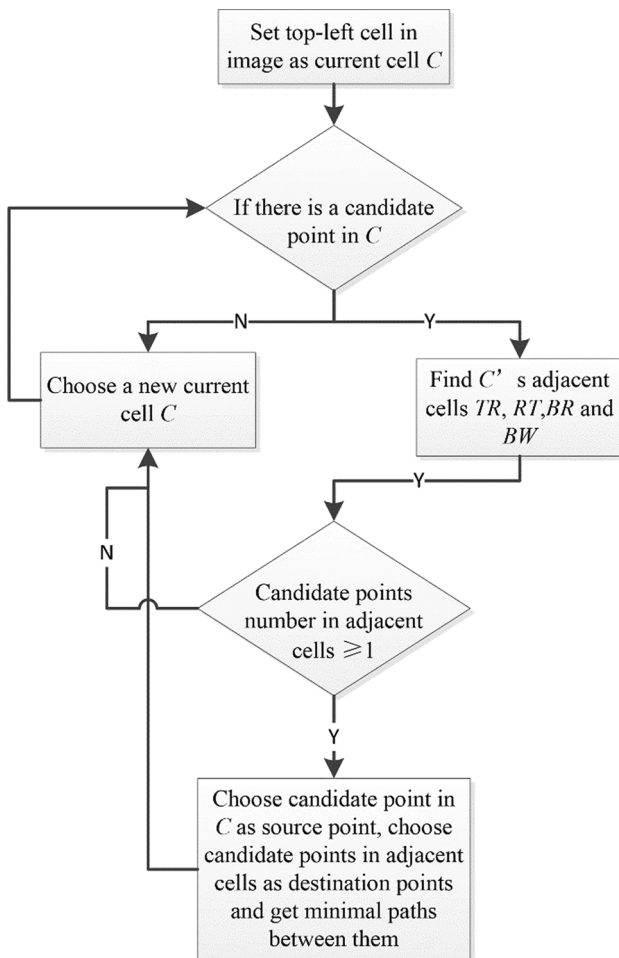


Fig. 4. Strategy of searching for source and destination candidate points.

There may be overlapping paths in the current MCPS, such as path (C, TR), which means the minimal path between the candidate point in cell C and that in cell TR, exactly equals path (C, RT) plus path (RT, TR). In this situation, path (C, TR) is removed to avoid subsequent redundant

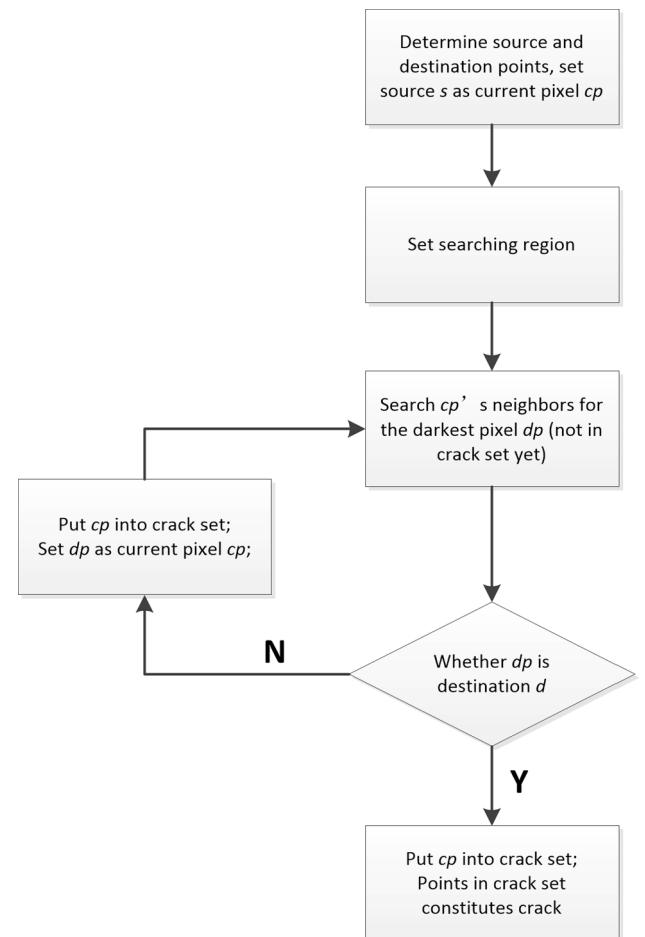


Fig. 5. Flow chart of improved Dijkstra algorithm.

calculations while path (C, RT) and path (RT, TR) remain.

However, this set now contains both the main part of the crack and some incorrectly detected crack paths outside the crack area. To remove the pseudocrack paths, a method consulting the threshold idea is proposed. The threshold T_r is set by the function below, and any paths with

a cost above the threshold T_r are removed from the MCPS,

$$T_r = \mu_r - k_r \sigma_r \quad (7)$$

where μ_r and σ_r are the mean cost and standard deviation of the MCPS, respectively, and k_r is an empirical coefficient.

In the main procedure of improved minimal path selection (IMPS), first, source and destination of the cost path are determined by the proposed strategy, and then, the minimal cost path is completed based on the improved Dijkstra algorithm. Next, overlapped paths are removed, and a threshold is used to remove incorrectly detected cracks to obtain the filtered MCPS, as shown in Fig. 6.

2.2.4. Postprocessing

The threshold T_r described above filters most of the incorrectly detected crack paths, but there are still three kinds of pseudopaths in the MCPS, which can be defined as pseudobranches, pseudorings and pseudo short cracks. The formation of pseudobranches occurs when a false candidate point detected outside the crack range is adjacent to the crack and connects to a candidate point on the crack with a minimal path, which represents a branch of the crack. Regarding pseudorings, one or more candidate points detected outside the crack range are adjacent to one or more candidate points on the crack, and they are connected to each other by a series of minimal paths to form a ring shape. A pseudo short crack is formed because of the connection between two or more incorrectly detected candidate points outside the crack, which is usually related to dark debris and dirt on the road surface. After all the steps, crack skeleton can be performed, and its true width can be obtained through width expansion.

(1). Removal of pseudorings

First, the ring structures in the MCPS are detected. Generally, the length between two bifurcation points of a pseudoring should not be greater than a specific value, which is set as the length of 3 cells. Then, the ring is broken from the bifurcation points to obtain two independent paths. Finally, the path with the higher cost is removed, and the one that remains belongs to the real crack.

(2). Removal of pseudobranches

When detecting branch structures, generally a pseudobranch is less than a certain length. The length threshold L_b is set as the length of 2 cells. If a branch matches the condition that its length is less than $L_b = 2P$ pixels and its cost is not among smallest two within the branches from the same bifurcation point, then it is eliminated as a pseudobranch.

(3). Removal of pseudo short cracks

First, short crack structures are detected. A gray threshold T_b is set by the following function:

$$T_b = \mu_b - k_b \sigma_b \quad (8)$$

where μ_b and σ_b are the mean cost and standard deviation of the MCPS after the removal of pseudobranches, respectively, and k_b is an empirical coefficient. If a short crack is shorter than L_b with a cost higher than T_b , then it is removed as a pseudo short crack.

(4). Width expansion

Since the minimal paths found in the previous steps are single-pixel wide, which is not true for the crack, the true width of the crack needs to be obtained through expansion. Width expansion refers to the principle of region growing algorithm, which means that the expanded width pixels must be adjacent to the existing crack pixels and satisfy certain gray level conditions. Thus, the threshold idea is also used as the judgment standard for width pixels in the width expansion session, and the threshold parameters are set as follows:

$$T_w = \mu_w - k_w \sigma_w \quad (9)$$

where μ_w and σ_w are the grayscale mean and standard deviation of candidate point sets, respectively, and k_w is an empirical coefficient used to determine the range of the gray level of width pixels, which can be adjusted to obtain the true width of the crack.

In summary, the MCPS filtered by T_r needs to be eliminated of different wrongly detected part and the true width of the crack is obtained by width expansion. After these postprocessing procedures, as shown in Fig. 7, the binary image of the crack region is finally determined.

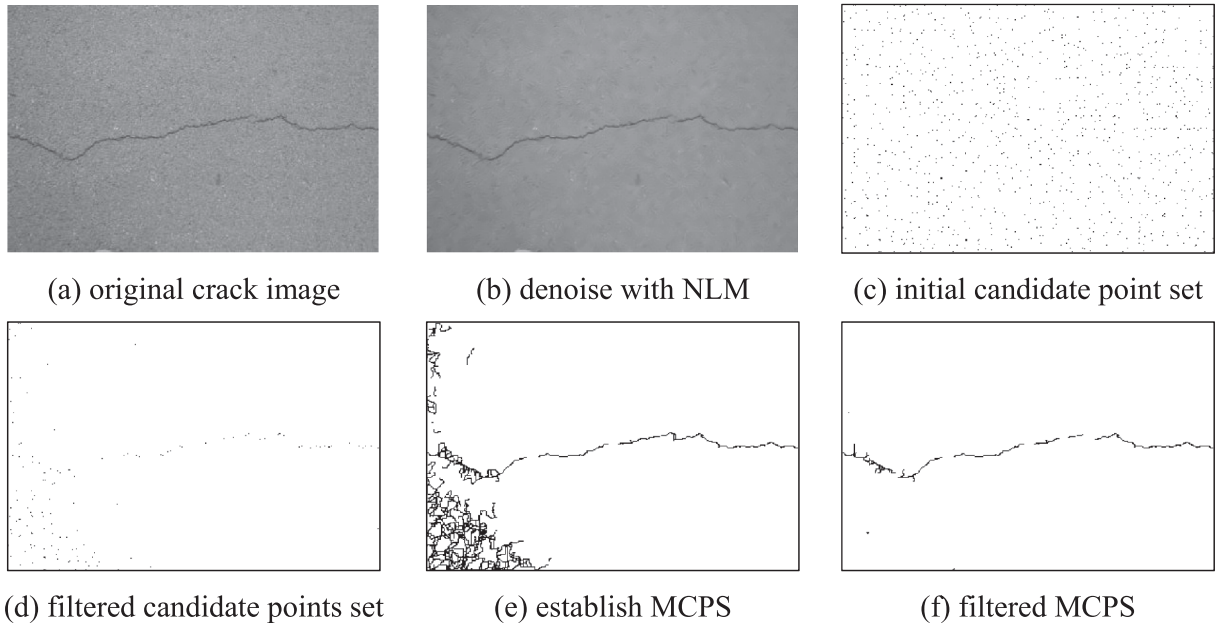


Fig. 6. Main steps of IMPS.

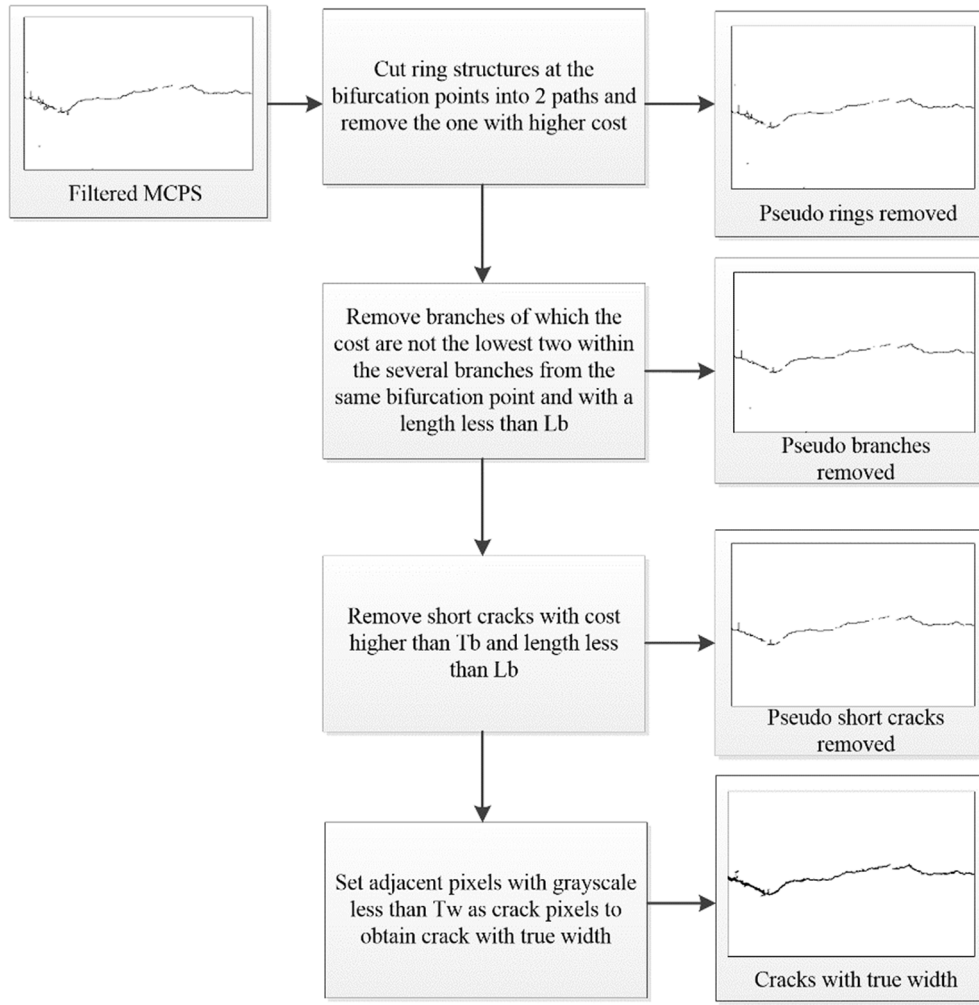


Fig. 7. Postprocessing procedure.

3. Results

The pavement distress images processed in this paper come from the CFD dataset [39]. The CFD dataset contains 118 RGB images with a resolution of 320×480 pixels. These images contain noise, such as shadows, oil spots, waterlogging, and nonuniform illumination. The PC used to process the distress images has an Intel Core i7-9750 3.60 GHz CPU and 8 GB RAM, and all the processes in this study are implemented using the MATLAB 2019b image processing toolbox unless otherwise noted.

The proposed algorithm based on the minimal cost path is elaborated in section 2.2. To verify the robustness of the proposed algorithm, crack images with different conditions, such as fine cracks, low resolution, shadows, and waterlogging, are applied to test the segmentation performance. The precision p , recall r , and F_1 score metrics are employed to quantitatively evaluate the segmentation performance. Precision p represents the ratio of the number of crack pixels correctly detected to the number of crack pixels identified. Recall r represents the ratio of the number of crack pixels correctly identified to the true number of all crack pixels. F_1 represents the harmonic mean of the precision and recall. These metrics obey the following equation:

$$\begin{aligned}
 p &= \frac{TP}{TP + FP} \\
 r &= \frac{TP}{TP + FN} \\
 F_1 &= \frac{2 \times p \times r}{p + r}
 \end{aligned} \quad (10)$$

The confusion matrix for binary classification is shown in Table 1.

To further compare the segmentation effects of different methods, Intersection over Union (IoU), an index that can well evaluate classification results, is used, as shown in Fig. 8.

Segmentation is essentially a pixel classification task. In this respect, IoU can better reflect the veracity of the model. There are 2 classes in crack segmentation: the crack class and the background class. For these 2 classes, Mean Intersection over Union (MIoU) can be expressed as:

$$MIoU = \frac{\frac{TN}{TN+FN+FP} + \frac{TP}{TP+FP+FN}}{2} \quad (11)$$

Some pictures were randomly selected as experimental pictures in Fig. 9. After many experiments, it is found that setting the parameters k_g ,

Table 1
Confusion matrix for binary classification.

Data	Classified as positive	Classified as negative
positive	True positive (TP)	False negative (FN)
negative	False positive (FP)	True negative (TN)

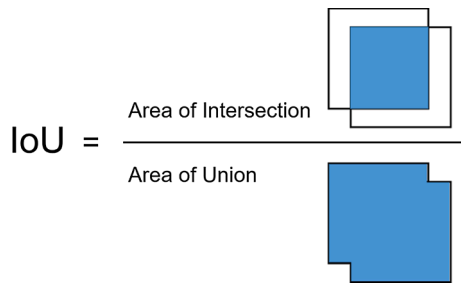


Fig. 8. Illustration of IoU.

k_r and k_w at 0.7, 0.2 and 4 can achieve a synthetic optimal segmentation result for the CFD database.

The other three methods are Otsu, region growing and edge detection. Otsu method is based on a threshold to split background and crack. For pavement images, the complexity of background and dirt may influence the performance of Otsu. Region growing requires to select one or a few target points and then searches neighborhood to collect similar points to get a target region. Edge detection uses mathematical morphology to get the margin of crack region and then fills the enclosed region to get crack.

It can be seen in Fig. 9 that the segmentation performance of IMPS is more reliable than that of other methods, and it can achieve an excellent segmentation performance both for fine cracks and alligator cracks. To quantitatively evaluate the segmentation performance of the proposed method, the p , r and F_1 metrics are estimated, as shown in Table 2. As shown in Fig. 9 and Table 2, the segmentation performance of edge detection method is the lowest among the adopted methods. Some pavement background texture pixels are also mistakenly detected as crack pixels. Otsu method tends to select scattered dark pixels in the background, which leads to inaccuracy. The effect of region growing method relies on the manually chosen points so it's unstable. What's more, manually selecting initial point(s) can be time-consuming and subjective. The proposed approach has the highest detection accuracy, with an F_1 score reaching 65.85%. This result shows that the segmentation performance of IMPS is better than the segmentation performances of other conventional methods and is more accurate than traditional MPS method.

As Fig. 10 shows, for these twelve images, the MIoU (Mean Intersection over Union) of IMPS reaches 73.99%, which is a rather excellent result that is far better than the results of the other tested segmentation algorithms. The MIoU of Otsu method, region growing and edge detection are 66.39%, 59.84% and 59.12%, respectively. The MIoU of traditional MPS reaches 70% but is still less than IMPS.

The advantage of IMPS is that it also improves the efficiency of MPS algorithm. To explain the promotion of efficiency more clearly and concretely, the improved algorithm is compared with MPS. Table 3 shows the computational performance of IMPS and MPS. According to Table 3, if the side length of cells is set as $P = 8$, the computational time is significantly reduced by approximately 56% when applying IMPS. The experimental results for all sizes show an average efficiency increase of approximately 50%. The result shows that IMPS performs better in terms of efficiency than conventional MPS method. Therefore, it is beneficial to adopt IMPS in the future.

4. Discussion

4.1. Influence of preprocessing session

To study the influence of preprocessing, the original twelve crack images used in the experiment described in Section 3 and the images after preprocessing are segmented with IMPS separately to extract cracks. The results are shown in the line charts below.

Fig. 11 (a) shows the precision of the twelve images under two

preprocessing conditions, and Fig. 11 (b) reflects the results of the recall index. It can be seen from the charts that after preprocessing, the resulting curves are more stable. Additionally, the average precision and recall without preprocessing session are 62.13% and 62.33%, respectively, while after preprocessing, they are 62.54% and 70.25%, respectively, which are both higher.

Fig. 12 shows the F_1 scores of the twelve images under two different preprocessing conditions. After nonlocal means filtering, the average F_1 score of the twelve images reaches 65.85%, while it only reaches 60.01% without preprocessing. It can be seen that preprocessing can certainly improve the segmentation results for most images, although for very few images (such as image 9), the results obtained without preprocessing are slightly better than those obtained after preprocessing.

For the twelve representative images, the standard deviation of F_1 obtained by using nonlocal means as the preprocessing algorithm is 0.0897, which is smaller than that without preprocessing procedure, 0.0986. This result proves that nonlocal means denoising can improve the stability of the proposed method and suppress the problem that the level of the extraction results of different types of crack images can be too large.

Therefore, it is proven that preprocessing can effectively remove some noise from images to improve the segmentation of the images.

Then, the two preprocessing algorithms, nonlocal means filtering and median filtering, are compared to determine the influence of the different denoising algorithms on the proposed method.

In the background of Fig. 13 (a), the original image, there are some dark pixels and stains with grayscale values similar to those of the crack. Although the median filtering algorithm reduces the background noise, the gray level of the crack is also reduced, and the image becomes blurred. The image after nonlocal means filtering not only has reduced background noise but also has a more prominent crack without a loss of clarity of the crack. Therefore, subjectively, the latter filtering algorithm is considered to be better.

The comparison of two preprocessing algorithms by using PSNR and SSIM indices mentioned previously is shown in Fig. 14. It is obvious that the PSNR and SSIM of nonlocal means filtering are both higher than those of median filtering, which means that nonlocal means filtering has a better effect on image denoising.

Note: MF and NLM represent median filtering and nonlocal means filtering, respectively.

It can be seen from Fig. 15 that the crack extraction results obtained by using nonlocal means as denoising algorithm are more accurate than those obtained by median filtering algorithm.

4.2. Influence of cell size

Here, we fix the values of parameters k_g , k_r and k_w as 1, 0 and 2.4, respectively. Then, different sized cells are used to conduct the experiment. The results are as follows.

In an image, the number of crack candidate points depends on the size of the cell. The smaller the cell size, the greater the number of crack candidate points and the greater the computing resources and time consumption for a single image. However, if the cell is too large, the number of crack candidate points may be too small and some candidate points belonging to cracks may be ignored, leading to decreased accuracy. Therefore, a reasonable cell size is needed to improve the computational efficiency on the premise of ensuring the accuracy.

Different sizes of cells ($P = 4, 8, 10, 16, 20, 40$) were used to determine the optimum cell size, and Table 4 gives the average time consumption of the twelve pavement images under different cell sizes. As seen in Table 4, although a larger cell can save computational time, the accuracy of the result is unsatisfactory because a large cell cannot avoid missing crack points. Small-size cells, such as those with a side length of 4, can certainly increase the F_1 of most pictures, but they require much more time, which leads to inefficiency. In general, when $P = 8$, the segmentation of an image can both be efficient and achieve

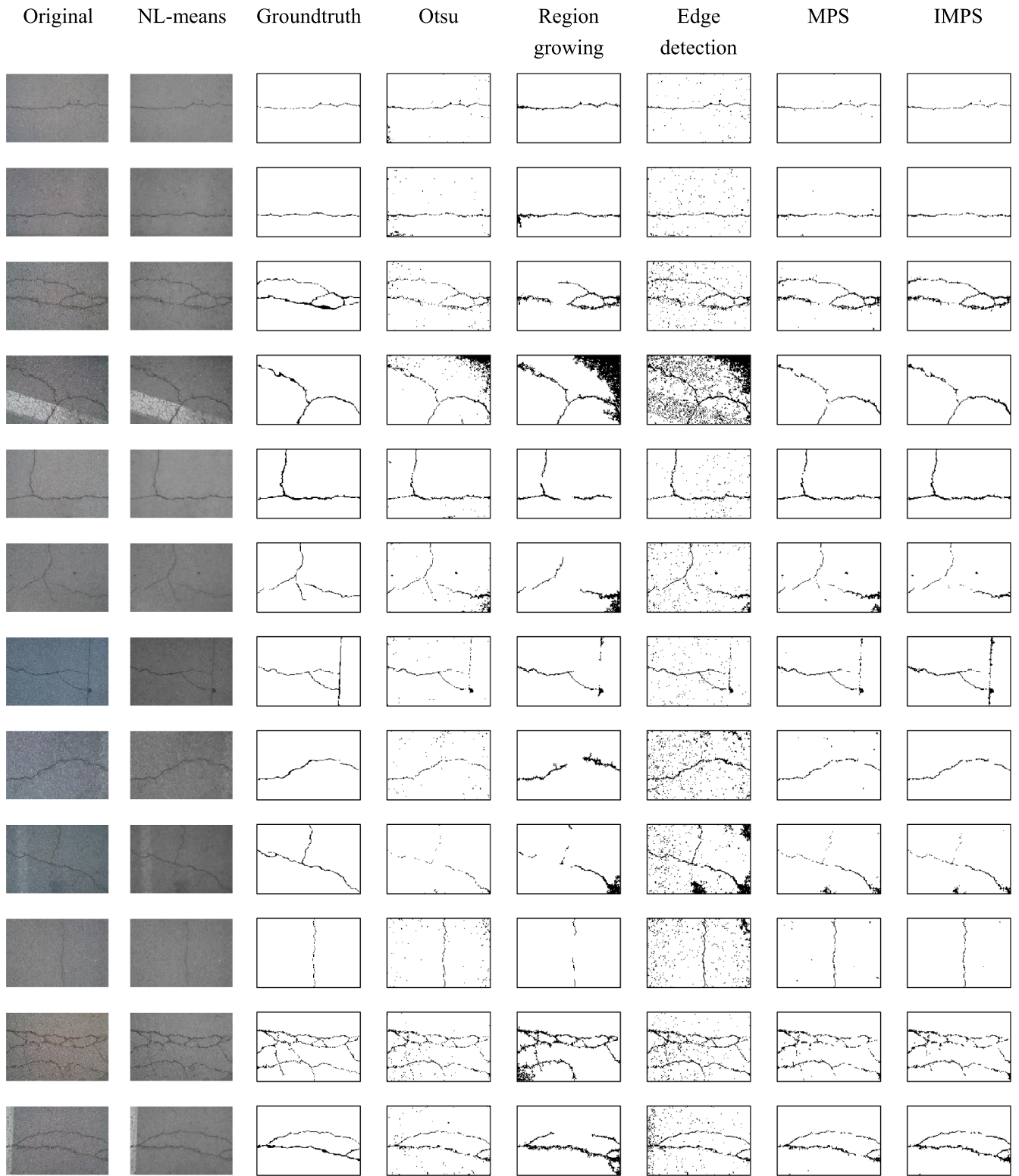


Fig. 9. Comparison of the segmentation performances of different methods.

rather satisfactory results. As a result, $P = 8$ is the selected side length in this paper.

4.3. Robustness analysis

To verify the robustness of the proposed method, crack images with different conditions such as low-resolution, thin crack, wide crack, crack

on concrete and stained images are randomly picked and processed by IMPS. Note that the parameters of IMPS in the process are kept fixed. Parameter n is set to 8, k_g to 0.7, k_r to 0.2, k_w to 4.

Testing images are randomly chosen from dataset DeepCrack [26] and CFD [39], Fig. 16 shows the segmentation results of the defect images mentioned above. Table 5 reveals that IMPS can achieve an overall excellent segmentation performance on crack images of different

Table 2
Crack detection performance evaluation.

Metrics	Otsu	Region growing	Edge detection	MPS	IMPS
P	44.92%	38.89%	25.78%	59.33%	62.54%
R	51.60%	70.00%	62.84%	64.20%	70.25%
F1	45.69%	46.53%	34.32%	62.04%	65.85%
MIoU	66.39%	59.84%	59.12%	71.30%	73.99%

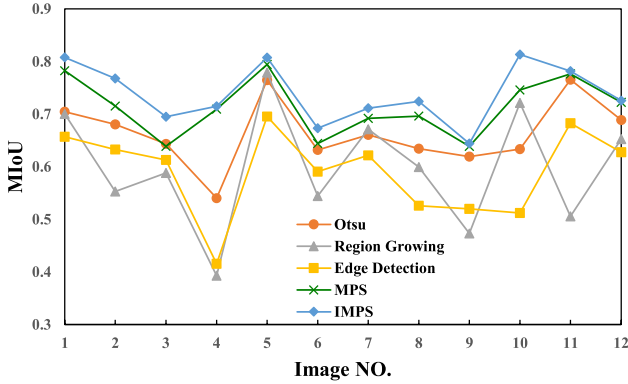


Fig. 10. Statistical chart of MIoU.

conditions, especially for cracks on concrete. For wide cracks (Fig. 16-6,7), although not all the crack part is darker than background, the extraction of crack pixels can be satisfactory. As for low-resolution images (Fig. 16-2,3), the boundary between the crack and the background is blurred so the edge is hard to be perfectly detected, but the F1 score

and MIoU is still at a good level. It can be seen in Fig. 16-4 that IMPS can get rid of most small stains but large dark stains may be left. This method has a good recognition accuracy for crack images with complex background (Fig. 16-5). The accuracy for images with stains is at a good level. As a result, the robustness of the proposed method is proved.

5. Future analysis

According to experiments above, the average processing time of a 480*320 size image by using general laptop is about 4 s, which is expected to reduce to around 1 s or even lesser if using a high performance workstation. As for deep learning method, the segmentation time of a 256*256 size image can be about 0.5 s [26], but the labeling section may cost 5 min for a linear crack, even 3 or 4 times longer for a crocodile crack, which is quite time-consuming. what's more, manually labeled

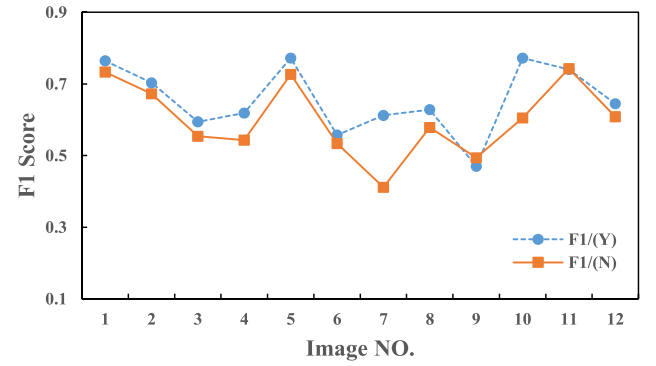
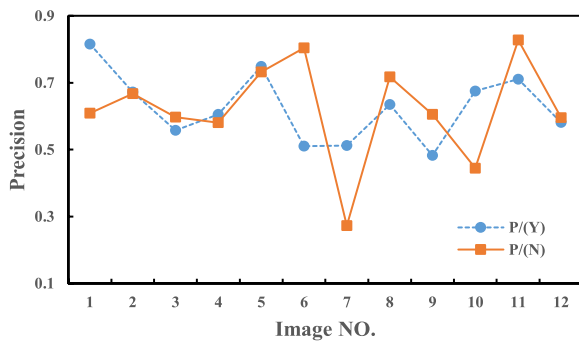


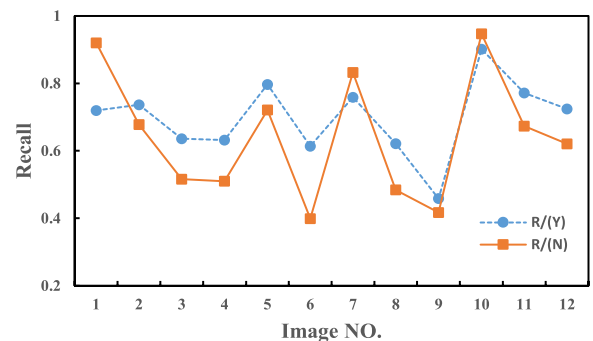
Fig. 12. F_1 scores under different preprocessing conditions.

Table 3
Computational time of IMPS and MPS (unit: s).

Cell size	4		8		10		16		20		40	
	IMPS	MPS	IMPS	MPS	IMPS	MPS	IMPS	MPS	IMPS	MPS	IMPS	MPS
1	3.68	7.05	1.70	2.96	1.44	2.54	1.46	2.78	2.22	3.44	1.36	2.78
2	12.47	13.45	2.52	5.63	2.15	4.34	2.06	3.86	1.68	3.38	1.38	1.91
3	18.83	21.10	4.15	8.95	3.81	8.20	3.49	6.41	3.23	5.97		
4	13.89	21.73	3.19	8.34	2.98	7.62	2.42	6.37	2.03	4.90	1.47	2.78
5	21.68	24.12	3.01	6.18	2.52	5.33	2.12	4.49	1.87	4.09	1.03	1.59
6	13.69	20.58	2.62	6.17	2.11	5.43	1.43	2.85	1.23	2.33		
7	8.93	13.67	2.26	5.39	2.26	4.31	1.64	3.60	1.48	2.55	1.18	2.22
8	10.29	16.99	2.82	7.64	2.30	5.85	1.76	4.33	1.45	3.40	1.10	1.81
9	16.03	27.15	3.99	10.30	3.26	9.10	1.81	4.81	1.27	2.83		
10	5.70	11.66	2.67	5.30	2.31	4.62	2.17	3.79	1.37	2.06	1.14	1.37
11	8.03	15.20	3.72	8.20	3.16	6.94	3.12	6.81	2.85	6.55	1.15	1.55
12	5.67	9.84	2.69	5.55	2.80	6.23	2.72	5.80	2.20	4.58	4.48	9.44
mean	11.63	17.04	2.96	6.75	2.65	6.03	2.20	4.64	1.93	3.88	1.20	2.12
Efficiency improvement	31.76%		56.23%		56.01%		52.53%		50.41%		43.47%	



(a) Precision p in two conditions



(b) Recall r in two conditions

Fig. 11. Comparison of the segmentation results whether using preprocessing.

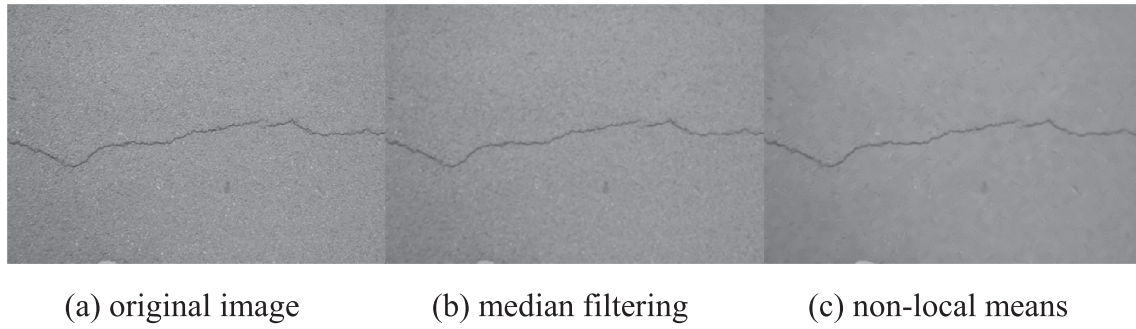


Fig. 13. Comparison of different denoising methods.

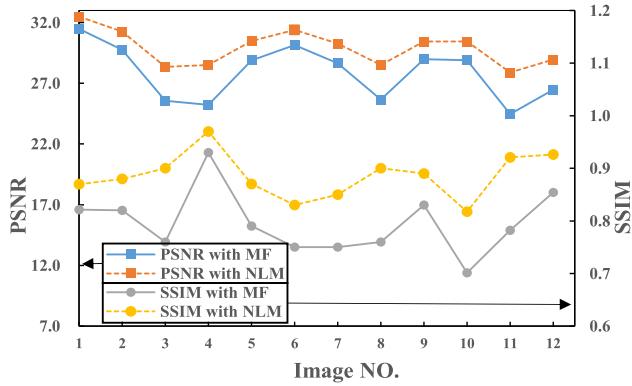


Fig. 14. Evaluation of the two preprocessing algorithms.

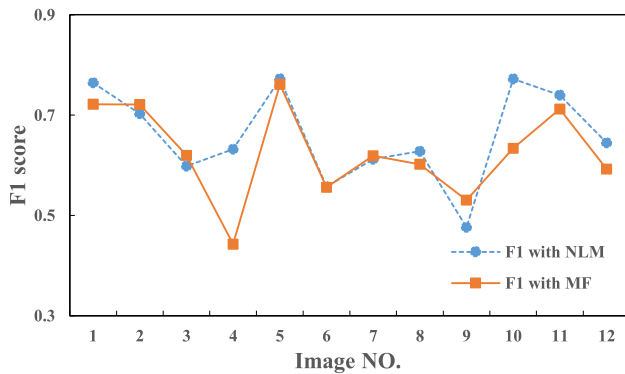


Fig. 15. Performance of the proposed method using NLM and MF separately.

Table 4
Time consumption of different cell sizes.

Cell size	4		8		10		16		20		40	
	Time	F1	Time	F1	Time	F1	Time	F1	Time	F1	Time	F1
1	3.68	0.7147	1.70	0.6964	1.44	0.5971	1.46	0.4636	2.22	0.5367	1.36	0.3622
2	12.47	0.6540	2.52	0.5986	2.15	0.5846	2.06	0.2425	1.68	0.3559	1.38	0.2617
3	18.83	0.4720	4.15	0.3922	3.81	0.3471	3.49	0.2597	3.23	0.1418		
4	13.89	0.4334	3.19	0.3464	2.98	0.3267	2.42	0.2920	2.03	0.2751	1.47	0.2347
5	21.68	0.6332	3.01	0.4211	2.52	0.3512	2.12	0.3662	1.87	0.2859	1.03	0.0441
6	13.69	0.5204	2.62	0.4518	2.11	0.3441	1.43	0.2516	1.23	0.0189		
7	8.93	0.4491	2.26	0.3211	2.26	0.3152	1.64	0.2153	1.48	0.2245	1.18	0.1106
8	10.29	0.5604	2.82	0.6509	2.30	0.5798	1.76	0.3599	1.45	0.3880	1.10	0.1497
9	16.03	0.4846	3.99	0.4413	3.26	0.4361	1.81	0.2801	1.27	0.1381		
10	5.70	0.3763	2.67	0.4257	2.31	0.3220	2.17	0.2760	1.37	0.2076	1.14	0.0957
11	8.03	0.6158	3.72	0.5123	3.16	0.4528	3.12	0.3327	2.85	0.3506	1.15	0.0842
12	5.67	0.6126	2.69	0.5528	2.80	0.5692	2.72	0.4548	2.20	0.3886	4.48	0.3080
average	11.63	0.5682	2.96	0.5110	2.65	0.4598	2.20	0.3205	1.93	0.2786	1.20	0.1447

Note: Blank cells in the table mean that the extracted crack is empty, which is the same in Table 4.

images may contain some subjective errors.

If the proposed method is introduced to the labeling section before deep learning algorithm, the labeling time will be greatly reduced and some errors caused by human's subjectivity can be avoided. This will also save a lot of manpower.

For future study, preliminary labeling results can be obtained by IMPS and manually corrected or supplemented to get accurately labeled images in a much shorter time. The labeled images can be used for training deep learning networks.

To reach a higher segmentation accuracy, there are 2 aspects for consideration:

- (1). The parameter setting should be further investigated. For crack images with different conditions (thin/wide crack, concrete/asphalt pavement background, light conditions, whether there are smudges/shadows, etc.), different combinations of parameter k_g , k_r and k_w are required to reach optimal segmentation results. More experiments are needed to set better parameter combination for crack images of different conditions. Several certain combinations of the parameters can be put forward and optimal segmentation results can be achieved by traversing these combinations.
- (2). The algorithm of removing three kinds of pseudocracks can be optimized. At present, there are still omissions and erroneous elimination when detecting and removing three kinds of pseudocracks. Thus a better algorithm of detecting and removing pseudo cracks can be further studied.

6. Conclusions

In this paper, some new strategies are proposed to optimize the traditional minimal path selection algorithm (MPS) to obtain improved minimal path selection (IMPS), which has a better crack segmentation

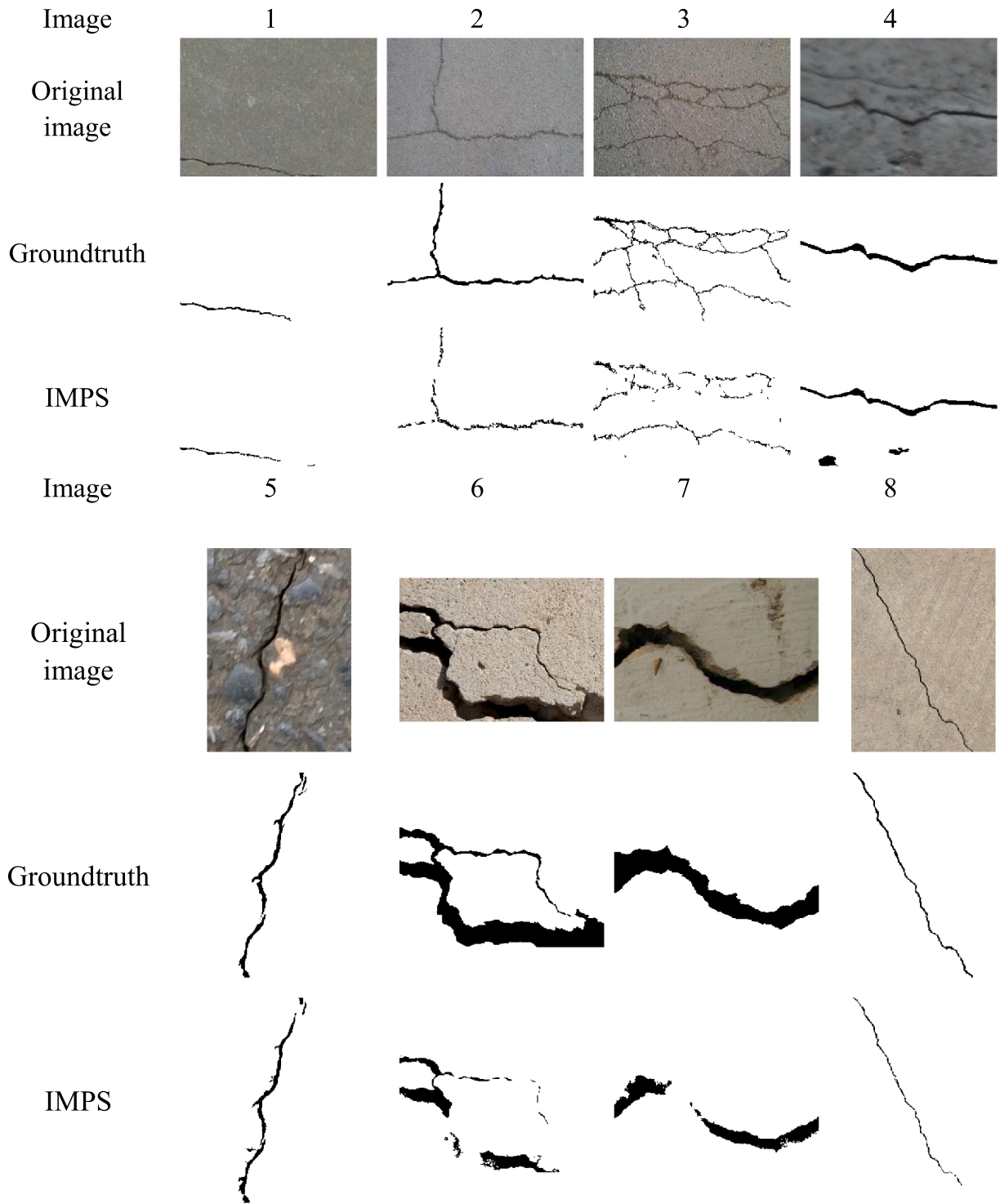


Fig. 16. Robustness analysis with crack images of different conditions.

Table 5
Statistical summary for robustness analysis.

	1	2	3	4	5	6	7	8	Average
P	92.27%	81.79%	87.61%	77.00%	90.52%	99.99%	99.97%	99.88%	91.13%
R	70.17%	52.22%	55.48%	84.38%	87.91%	35.07%	51.92%	56.62%	61.72%
F1	79.72%	63.74%	67.93%	80.52%	89.20%	51.93%	68.34%	72.28%	71.71%
MIoU	83.01%	72.53%	74.56%	82.99%	89.93%	61.12%	72.19%	77.98%	76.79%

performance than other traditional crack segmentation methods. The following conclusions can be obtained.

- (1). Novel cost path searching strategies are proposed to settle the problem of determining search regions and selecting source and destination points. The proposed strategy solves the problem of the low efficiency of the traditional minimal cost path calculation, and the efficiency is increased by approximately 56%. The accuracy of IMPS is also increased comparing with MPS.
- (2). The improved minimal cost path selection has a much better segmentation accuracy than Otsu method, region growing, edge detection and MPS. In terms of F_1 metric, the accuracy of IMPS is 20% higher than that of Otsu method and region growing method and is 30% higher than that of edge detection method.
- (3). The importance of denoising the original image is proven, and the nonlocal means algorithm is used to denoise pavement images. Two metrics, PSNR and SSIM, are used to evaluate the performance of different types of filtering algorithms. The results show that nonlocal means filtering performs better than other algorithms, such as median filtering. The cell size is also carefully chosen through experiments to balance efficiency and accuracy.
- (4). Using IMPS for labelling crack images to prepare training materials for deep learning methods.

However, it is undeniable that although the improved method reduces the computational time and has a rather excellent segmentation performance, wrong detection happens as well. The ability of this method to distinguish road smudges or shadows still needs to be improved. In the future, we will improve the robustness and generalization performance of proposed method to segment crack images with more complex backgrounds.

CRedit authorship contribution statement

Yizheng Chen: Methodology, Software, Writing - original draft, Writing - review & editing. **Jia Liang:** Methodology, Writing - review & editing. **Xingyu Gu:** Methodology, Writing - original draft, Writing - review & editing. **Qipeng Zhang:** Visualization. **Hanyu Deng:** Methodology. **Shuwei Li:** Methodology, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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